

11/6/25

Day 3 (7+8)

SQL is

Data Structure in SQL

⇒ First we can create a database, inside DB we can create 'n' number of table. All these tables interrelated to each other. It store data inside the table. Each table in a database will have a own set of data.

⇒ Table is the combination of Rows & Columns. Column will tell us the general structure.

⇒ In a db design a structure is called Schema.

⇒ Rows talk about individual data.

⇒ Column are also known as Attribute or Field

⇒ Rows are also known as Records or Tuples

Creating our First DB:-

⇒ create CREATE - is used to create db & objects

Syntax:

; → it is full stop

CREATE Database_databasename;

ex:

CREATE database School;

Before create table

⇒ USE keyword is used to select the db.

Syntax:

USE database School;

note: we can't create ^{two} database with same name.

How to create a table:- ~~(X)~~ Constraints is optional

Syntax:

Create table table-name (col name1 datatype (constraints),
col name2 datatype constraints, upto
col name n datatype constraints);
Number - int
Name - Varchar img, video, blob

ex:

Create table Student (Student ID ^{int} ~~char~~ unique, Student name
String Varchar);

Note

⇒ Col name & Datatypes are mandatory, constraints is optional.

⇒ They can't be ^{two tables in} same name in same database

Activity: Database create
↓
Table create →

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Day 4 (7+08)

Database Related Syntax

1) → To see list of database

Syntax:

Show databases

2) → To create a database

Syntax:

Create database database name;

ex:

Create database School;

3) → To Select a database

Syntax: Use name

ex: Use School

Syntax: Select database ();

Table Related Syntax :-

⇒ create table student (Col1 datatype constraint, Coln ...)

⇒ See list of table

Syntax :

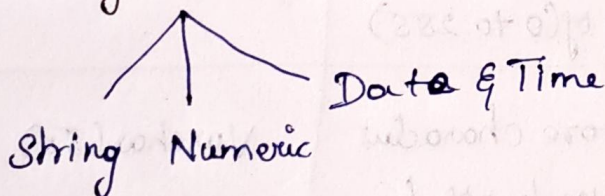
Show tables

⇒ Table detail

Syntax: describe table tablename

describe tablename

Data types in SQL :-



String

" "

Numeric

int, float

⇒ Each column in a db table, it compulsory required to have a name & datatype.

⇒ Datatype defined in

⇒ SQL developer must decide what type of data that will be stored ^{inside} each column while creating table.

⇒ The Datatype is a data guideline for SQL to understand what type of data is expected inside of each column.

⇒ SQL datatype are classified into

- 1) String (" ")
- 2) Numeric (Int, float)
- 3) Date & time

String datatype:-

Datatype	Description	Usage
Char(Size)	Can store character of fixed length/ String of (0 to 255)	Char(50)
Varchar(Size)	Can store characters of given length/ String (65535)	Varchar(50)
Blob(Size) (img, video)	To store large object file	Blob(1000)